

The idea to set up an educational platform first came to Subhajeet Mukherjee back in 2018, when he was studying in college. He says, “Many students in their initial months at the college were being taught programming through colourful drag and drop tools without any emphasis on operating systems and how these work. Programming in the real world is not as simple as drag and drop.”

The Programming Foundation (TPF) focuses on computer science education that is based on written instructions and interactive experiences. It requires no app download, no sign-ups, and no registration or the need to share any personal information. It provides volunteers with hands-on experience, enabling them to work together as a team by developing open source tools to improve the platform.

Google Ad Grants was a huge help

TPF was founded in February 2020. Theodore Rolle joined it as the secretary and technical advisor, post which the platform was launched in May, during the pandemic. Mukherjee says, “We never thought we’d be able to survive in 2020. We were fortunate enough to receive the Google Ad Grants along with other in-kind support from tech companies.”

The foundation’s goal is to spread the widespread adoption of a new and improved curriculum on computer programming and operating systems, and for this knowledge to become mainstream in the general population.

Why did he decide to work with open source? Mukherjee says, “A seamless experience is the need of the hour. It was a well-thought out decision to prioritise learning and interactive experiences over profits. Adopting a non-profit model along

with a commitment towards open source seemed more sensible to us.”

No sign-up option

Mukherjee adds that TPF is committed to open source, and plans on developing and involving volunteers in building industry standard technologies and applications that solve real-world problems and make the world a better place.

But there are other platforms that teach similar things. Why should one opt for The Programming Foundation? Mukherjee says that a lot of the other learning platforms ask people to sign-up or even try to charge them after one or two lessons. Many of them have almost no emphasis on operating systems and are unable to provide volunteers with hands-on experience. Instead, there is a lot of emphasis on Web development, and making people think that HTML and CSS are programming languages.

Mukherjee adds, “We don’t have any of that. We don’t even have a sign-up option. People can directly learn from the platform, and volunteers can join TPF to improve the platform. Our goal is to build a smarter general population so that no one is left behind.”

TPF is a tax-exempt non-profit. It solely relies on donations and grants. Mukherjee says that receiving donations and grants remains the biggest challenge. Grants are important to accommodate legal fees, general expenses, and hire full-time engineers for consistent development.

The most visitors are from the US, India and the UK

At present, the three most important domains for learning on TPF, based on analytics, are UNIX, Python and C. The top three countries, based on visitors, are the United States, India and the United Kingdom. The

platform has had 28,000 page views and attracted more than 10,000 users and learners.

So, where does he see TPF in the future? Mukherjee says, “I hate to say this but, generally, non-profits are technically incompetent. TPF’s plans are highly ambitious. We’ve already begun research towards making the foundation a platform that can be accessed by differently-abled people without the use of any dedicated/ assistive devices.”

Work is on for building an operating system that runs on the open instruction set architecture RISC-V. The goal is to tap into regions where the Internet is still inaccessible to people due to challenging economic situations and infrastructure. The long-term plan is to reduce the cost of a normal computing device and develop a native, singular and rigid interface similar to the *Learn* page of the TPF platform. Mukherjee feels that working with the government to achieve milestones will be very important for TPF.

Open source is important for a better and collaborative future

As someone working in this space, what does Mukherjee think about the state of open source in India? He says, “I’m fortunate enough to be part of the two greatest countries in the world, India being my homeland and the US being my second home — at least I consider it that way. I expect a lot of more open source initiatives coming out from India. Similarly, I think there’s a lot we can do to improve research on open source in the US. Defence based R&D agencies, governments, institutions, and think tanks can contribute to and benefit heavily from open source projects.”

Mukherjee feels that open source is important for a better and collaborative future. And that government and research institutions should adopt open source research and publication, which will significantly increase innovation in many fields. 